

PROJECT DEEP DOCUMENTATION

Student / User Management Web Application

Project Overview

This web application is designed to manage students and users using a structured backend architecture. It focuses on secure authentication, centralized database management, and modular PHP files to ensure scalability, clarity, and maintainability.

Technology Stack

The project is built using Core PHP for backend logic, PDO for database communication, MySQL for data storage, and Apache server through XAMPP. The system runs in a local development environment.

Project Architecture

The entire project follows a modular architecture where configuration, authentication, logic, and presentation are separated. This improves debugging, security, and future expansion.

Database Connection Layer

A dedicated database configuration file manages the connection with MySQL using PDO. All other files depend on this connection, ensuring a single source of truth for database access.

Authentication System

The authentication system is session-based. After successful login, session variables are created to track authorized users across protected pages.

Session Protection

Every restricted page verifies session availability before granting access. Unauthorized users are redirected to the login page automatically.

Student Management Module

This module handles student data entry, storage, retrieval, and display. Each responsibility is handled by separate files to keep logic clean.

Attendance Handling

Attendance-related files manage insertion and retrieval of attendance records. Errors are caught using PDO exception handling, ensuring application stability.

Logout Mechanism

The logout process safely destroys session data and redirects the user back to the login page.

Security Practices

The project uses prepared statements, centralized database access, controlled sessions, and structured error handling to maintain security.

Conclusion

This project demonstrates a backend-focused, secure, and scalable PHP application design suitable for academic submission and real-world use.